Simulation Arena

# Approval Bottleneck: Where Deals Actually Get Stuck

**CF Approval is the jam: 87.4% busy, adding 3.4 days of waiting, and it is why 50.7% of requests miss the 10-day promise. $3.02M of revenue is stuck behind it. Auto-approve requests under 10% would release $2.65M of that.**

Seed 42 | Run run_43bf44989cf1 | 4,000 scenarios tested | Discrete-event queueing simulation (arr… | 21 Sep 2026, 21:01

Same seed, same numbers - this run can be reproduced exactly.

## How this works

We rebuilt your approval process as a queue. Requests arrive the way yours do, each waits for a free approver, gets worked on for a realistic amount of time, then moves to the next step - and a request that waits too long sometimes gives up and walks. Running thousands of requests through that queue shows where the pile-up forms, how long the whole journey takes, and how many requests would still be sitting there when the quarter closes. We then re-ran the same queue with each fix in place to price them against each other. Every option is run on the same dice - same arrivals, same requests - so the difference between two options is the fix itself rather than luck.

## The numbers that matter

HOW LONG A REQUEST TAKES TODAY
**10.1 days**
Half of all requests clear in 10.1 days or less.
vs promise is 10 days

CLEARED INSIDE THE 10-DAY PROMISE
**49.3%**
The share of requests that still land in the quarter they were meant to.
vs 50.7% miss it

WHERE IT JAMS
**CF Approval**
This step is 87.4% busy and adds 3.4 days of waiting - the single biggest cause of the delay.

BUSIEST STEP IS BUSY
**87.4%**
Above about 85% a queue stops clearing: any hiccup becomes a backlog.
vs 3.4 days of waiting

REVENUE STUCK BEHIND THE QUEUE
**$3.02M**
Deal value of the requests that miss the quarter, times the 15.0% you told us is lost when a deal slips.
vs $38.36M of deals pass through in a quarter

WALK-AWAY RATE
**0.0%**
Requests that waited more than 25 days and gave up.

BEST FIX: AUTO-APPROVE REQUESTS UNDER 10%
**-3.5 days**
Median time goes from 10.1 to 6.6 days and requests inside the promise go from 49.3% to 94.0%.
vs $2.65M of revenue released

## How we got there

### 1. The process we rebuilt

10.2 requests a week arrive and pass through Deal Desk Review -> CF Approval -> Legal, with 6, 4, 5 approvers respectively. Measured from 260 requests in your own history, spanning 179 days.

Evidence: $38.36M of deals through the queue per quarter

### 2. Where it jams

CF Approval is 87.4% busy and makes requests wait 3.4 days on average - 33.2% of the whole journey is spent sitting in front of that one step.

Evidence: 87.4% busy, 3.4 days waiting

**3. What that costs**

Only 49.3% of requests clear inside the 10-day promise, so 50.7% slip out of the quarter. At 15.0% of deal value lost per slip, that is $3.02M of revenue at risk. Nothing gives up outright, but the slowest tenth wait 17 days.

Evidence: $3.02M at risk

**4. What the fixes are worth**

Auto-approve requests under 10% is the best of them: median time falls from 10.1 to 6.6 days, requests inside the promise rise from 49.3% to 94.0%, and $2.65M of revenue comes back into the quarter.

Evidence: $2.65M released

**5. The bottom line**

The constraint is not effort, it is capacity at CF Approval: the work itself takes 2.7 days, the waiting takes 3.4. Add capacity or take work out of that queue - asking people to work faster will not move the number.

Evidence: 2.7 days work, 3.4 days waiting

## Charts

### How long requests actually take

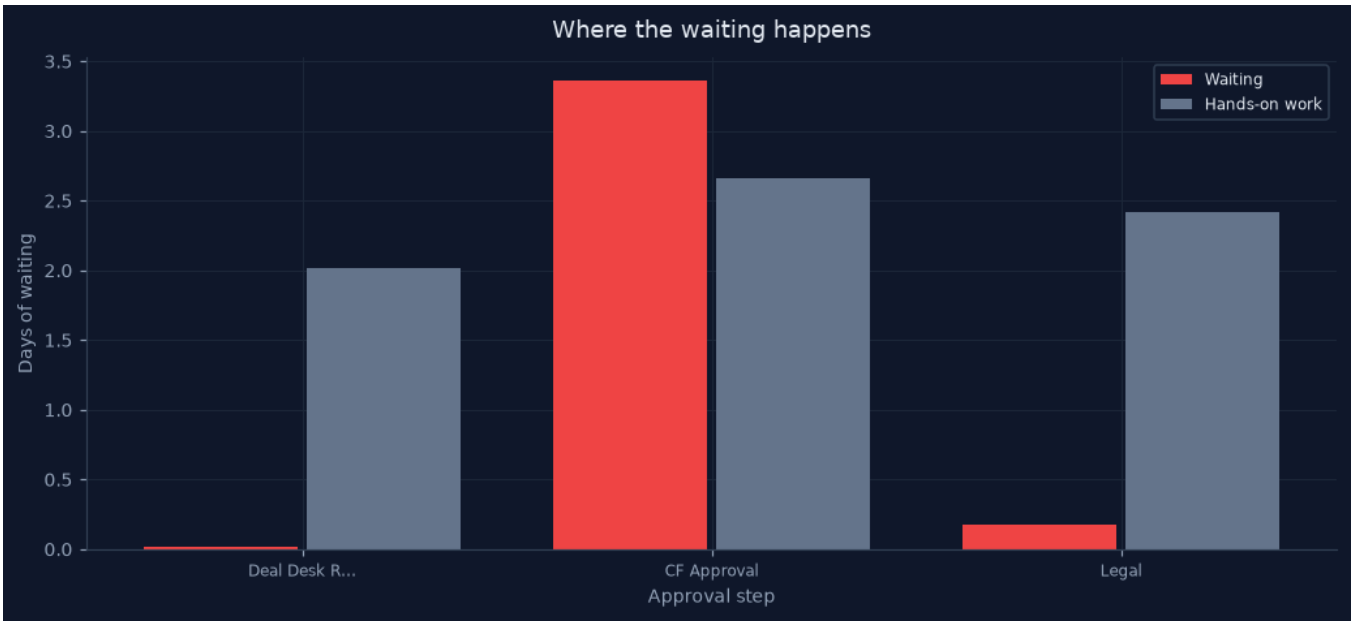
Today's process; the promise is 10 days




*What this shows: Everything to the right of the promise line is a deal that slips out of the quarter. The long tail on the right is the queue, not the work.*

### Where the waiting happens

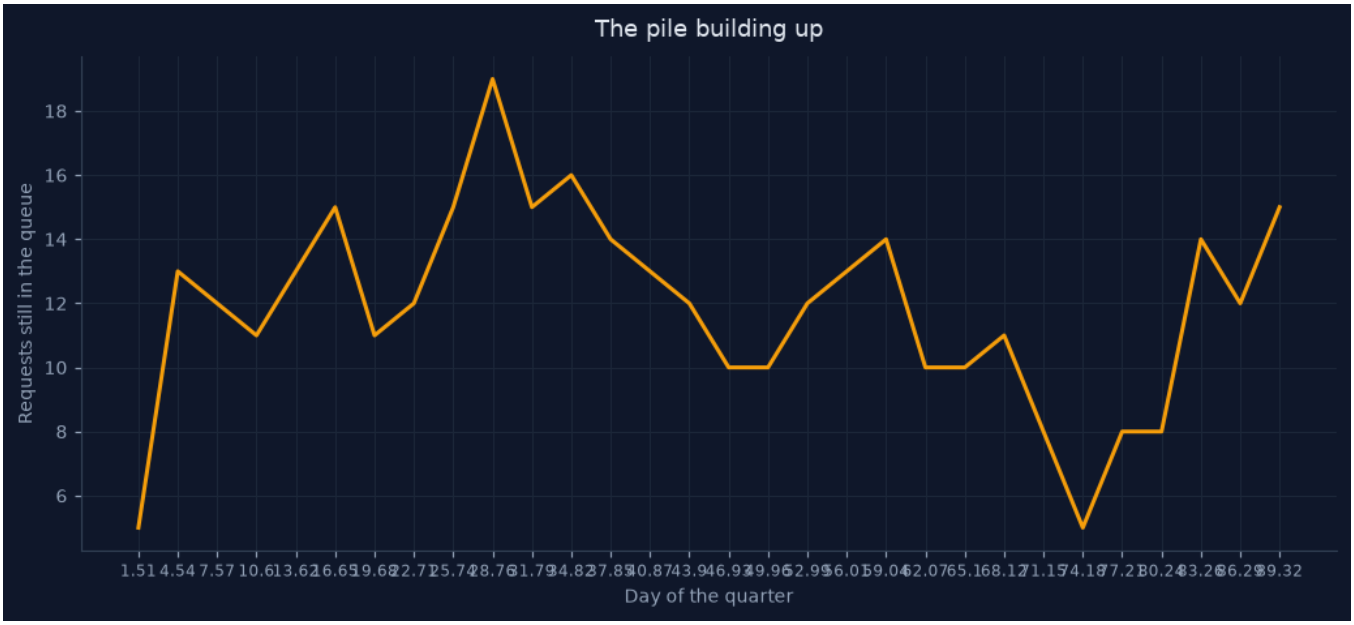
Average days a request sits before each step even starts




*What this shows: The tall red bar is the bottleneck. Grey is the actual work - if red dwarfs grey, the problem is capacity, not effort.*

### The pile building up

Requests open at any point in the quarter




*What this shows: A line that keeps climbing means the queue never catches up - that is the definition of a bottleneck.*

## Tables

### Each step, measured

Waiting is the queue; hands-on is the actual work. 'Days in your file' is the door-to-door time you recorded - waiting and work together - which is why it is bigger than hands-on.

| Step | Approvers | Requests | Days in your file | Avg wait (days) | Slowest 10% wait | Longest wait | Hands-on (days) | How busy | Share of all waiting | Column 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| Deal Desk Review | 6 | 649 | 2.1 | 0.0 | 0.0 | 0.9 | 2.0 | 47.6% | 0.6% | - |
| CF Approval | 4 | 649 | 7.6 | 3.4 | 6.5 | 15.3 | 2.7 | 87.4% | 94.2% | BOTTLENECK |
| Legal | 5 | 649 | 2.7 | 0.2 | 0.7 | 2.5 | 2.4 | 60.7% | 5.2% | - |

The bottleneck step is the one with the most waiting per unit of busyness - it is the reason the quarter slips.

### What each fix is worth

Revenue at risk = deal value of the requests that miss the 10-day promise, times 15.0%.

| Option | Days to clear (median) | Slowest 10% (days) | Inside the promise | Walk-away rate | Busiest step busy | Revenue at risk | Gain |
|---|---|---|---|---|---|---|---|
| Today's process | 10.1 | 16.7 | 49.3% | 0.0% | 87.4% | $3.02M | - |
| One more approver in CF Approval | 8.1 | 11.3 | 77.2% | 0.0% | 73.1% | $1.40M | $1.62M released |
| Auto-approve requests under 10% | 6.6 | 9.3 | 94.0% | 0.0% | 64.1% | $366k | $2.65M released |
| One more approver in Legal | 9.8 | 16.5 | 51.9% | 0.0% | 87.4% | $2.85M | $168k released |

The highlighted row is the cheapest fix that releases the most revenue.

### The ten worst-hit requests

The requests that lost the most time in today's process.

| Request | Arrived (day) | Days to clear | Of which waiting | Stuck behind | Deal value | Value at risk | Outcome |
|---|---|---|---|---|---|---|---|
| Request 64 | 39 | 24.1 | 8.1 | CF Approval | $349k | $52k | Late |
| Request 104 | 61 | 22.1 | 13.5 | CF Approval | $743k | $111k | Late |
| Request 135 | 86 | 21.5 | 10.3 | CF Approval | $230k | $34k | Late |
| Request 103 | 60 | 21.4 | 15.4 | CF Approval | $391k | $59k | Late |
| Request 100 | 59 | 21.4 | 14.5 | CF Approval | $717k | $108k | Late |
| Request 133 | 83 | 21.3 | 11.7 | CF Approval | $788k | $118k | Late |
| Request 105 | 61 | 20.9 | 14.8 | CF Approval | $171k | $26k | Late |
| Request 81 | 49 | 20.6 | 10.3 | CF Approval | $235k | $35k | Late |
| Request 96 | 58 | 20.5 | 11.4 | CF Approval | $81k | $12k | Late |
| Request 97 | 58 | 20.5 | 12.1 | CF Approval | $209k | $31k | Late |

Multiply this pattern across the quarter and you have the size of the problem.

## What we recommend

**Fix CF Approval first. Auto-approve requests under 10% releases $2.65M of revenue and takes 3.5 days out of the process.**

**Risk Moderate** 51 out of 100 | Confidence: High

### Why

- CF Approval runs at 87.4% busy and adds 3.4 days of waiting - more than any other step.
- Only 49.3% of requests clear inside the 10-day promise, putting $3.02M of revenue at risk.
- The work in that step takes 2.7 days; the waiting takes 3.4. It is a capacity problem, not an effort problem.

### What to do

**1.** Auto-approve requests under 10% - worth $2.65M of revenue released and 3.5 days off the median.

**2.** Auto-approve anything under 10% discount without a CF Approval stop - that alone removes 40.0% of the queue.

**3.** Publish a daily view of the CF Approval queue with a 10-day clock on each request, so slippage is visible before the quarter closes.

**4.** Watch the tail: the slowest requests take 17 days. Nothing is walking away yet, but at that length the customer has already re-planned without you.

**The trade-off:** Adding an approver costs salary; auto-approving below a threshold costs a little margin on small deals. Both are far cheaper than the $3.02M currently at risk.

## What we assumed and what to watch

### Worth knowing before you act

• Fewer than half of requests clear inside the 10-day promise. Until the bottleneck is fixed, the forecast for this quarter is optimistic.

### Assumptions

• Arrivals and timing measured from your own approval history: about 10.2 requests a week, 3 steps.

• Hands-on time per step: Deal Desk Review about 2.1 days; CF Approval about 2.7 days; Legal about 2.4 days

• Where your file records door-to-door time (queue included), we backed out the real hands-on time by finding the working time that reproduces the waiting you actually saw, given how many approvers you have.

• Approvers per step: Deal Desk Review 6, CF Approval 4, Legal 5 (counted from the distinct names in your file)

• A request that waits more than 25 days gives up with 25.0% probability.

• Revenue at risk = 15.0% of the deal value of every request that misses the 10-day promise; average deal value $288,286.

• Every request is modelled as passing through every step; steps that only some requests reach therefore look slightly busier than they are.

• 5 simulated quarters of 200 requests each per option (1,000 requests in total); about 129 arrive and clear inside each 90-day quarter. Seed 42, so re-running gives identical numbers. Every option is tested against the same simulated requests, so the comparison is fair. Revenue at risk is the average per quarter.

• Request / ticket ID -> your 'request_id' column

• Day it was submitted -> your 'submitted_day' column

• Approval step -> your 'stage' column

• Days it spent in that step -> your 'days_in_stage' column

• Who approved it -> your 'approver' column

• Deal value -> your 'deal_value' column

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.01s | prepare | You chose 'stage' for 'Approval step'. |
| 0.01s | prepare | You chose 'days_in_stage' for 'Days it spent in that step'. |
| 0.01s | prepare | You chose 'approver' for 'Who approved it'. |
| 0.01s | prepare | You chose 'deal_value' for 'Deal value'. |
| 0.01s | prepare | Request / ticket ID -> your 'request_id' column |
| 0.01s | prepare | Day it was submitted -> your 'submitted_day' column |
| 0.01s | prepare | Approval step -> your 'stage' column |
| 0.01s | prepare | Days it spent in that step -> your 'days_in_stage' column |
| 0.01s | prepare | Who approved it -> your 'approver' column |
| 0.01s | prepare | Deal value -> your 'deal_value' column |
| 0.22s | prepare | Read 709 records into the model. |
| 0.22s | simulate | Running the simulation… |
| 0.35s | read | Reading how approval requests arrive and how long each step takes... |

| Time | Stage | What happened |
|---|---|---|
| 0.65s | prepare | 'Deal Desk Review': 260 requests, 6 approver(s), 2.1 days door to door - about 2.1 of that is real work. |
| 0.75s | prepare | 'CF Approval': 260 requests, 4 approver(s), 7.6 days door to door - about 2.7 of that is real work. |
| 0.84s | prepare | 'Legal': 189 requests, 5 approver(s), 2.7 days door to door - about 2.4 of that is real work. |
| 0.84s | prepare | Learned the process from 709 request rows covering 179 days - about 1.45 requests a day. |
| 0.84s | prepare | Rebuilding the queue: 10.2 requests a week, 6 in Deal Desk Review, 4 in CF Approval, 5 in Legal. |
| 0.84s | simulate | Running 1,000 requests through the queue (5 simulated quarters)... |
| 0.87s | summarize | Found it: CF Approval is the jam - 87.4% busy and 3.4 days of waiting. |
| 0.90s | simulate | Tested 'One more approver in CF Approval'. |
| 0.95s | simulate | Tested 'Auto-approve requests under 10%'. |
| 0.99s | simulate | Tested 'One more approver in Legal'. |
| 0.99s | summarize | Comparing the fixes... |
| 1.00s | summarize | Done. CF Approval is the bottleneck - $3.02M of revenue is stuck behind it. |